Complex Disordered Systems

Maier-Saupe Theory, Lebwohl-Lasher Model, Frank Elasticity, Topological Defects

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Aside (optional): constrained optimisation with Lagrange multipliers

Many physics problems can be cast in terms of constrained optimisation

Area	Problem	Constraint
Mechanics	Minimize potential energy of a pendulum	Fixed length of the string
Thermodynamics	Maximize entropy S	Fixed total energy E
Statistical Mechanics	Maximize Gibbs entropy $-\sum_i p_i \ln p_i$	Normalisation $\sum_i p_i = 1$, fixed $\langle E \rangle$
Liquid Crystals	Minimize Frank energy	Director normalization $\|\mathbf{n}\| = 1$
Liquid Crystals	Minimize Maier-Saupe free energy $f[p]$	Normalisation of $p(\theta)$

Method: Use **Lagrange multipliers** λ to convert constrained into unconstrained problem:

$$\mathcal{L} = f(\mathbf{x}) - \lambda g(\mathbf{x})$$

then solve $\nabla \mathcal{L} = 0$.

Aside (optional): geometrical understanding of Lagrange multipliers

- Take a function $f(x, y)$.
- A constraint is expressed by fixing the value of a second function $g(x, y) = c$
- We want to find the extrema of $f(x, y)$ subject to the constraint $g(x, y) = c$: It has to be a curve
- This means find the extremal values of f on the curve $\mathcal{C} : g(x, y) = c$
- We could parametrise $\mathcal{C}$ but we can also simply think in terms of its geometrical features.

Aside (optional): geometrical understanding of Lagrange multipliers

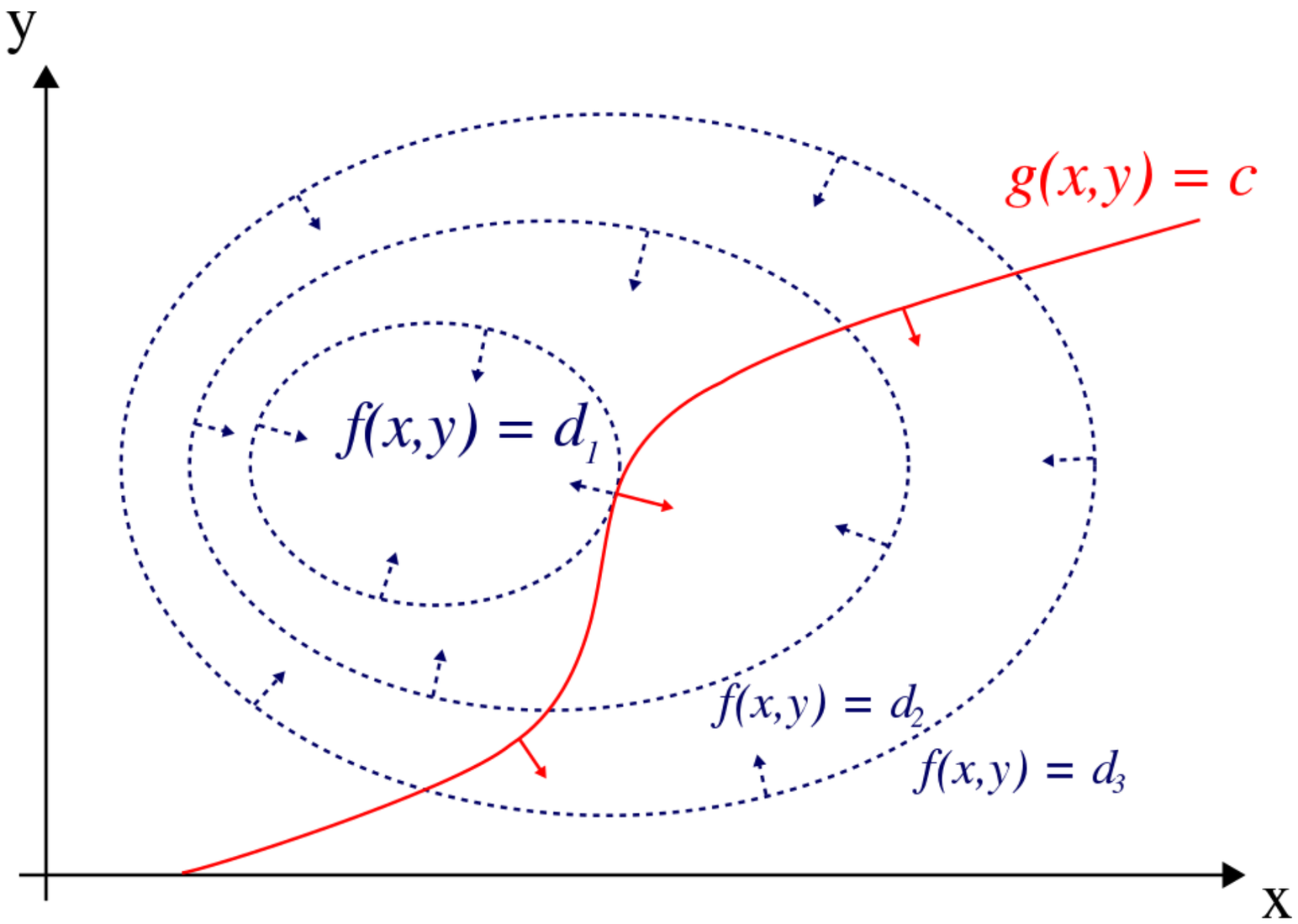
Take the example of a minimum:

- at every point of $\mathcal{C}$ there is a tangent vector $\vec{t}$
- we want to find the points on $\mathcal{C}$ where f is minimal. This means that *changes of f* along $\mathcal{C}$ are minimal.
- we can calculate such changes as the projection of the local gradient with the tangent vector and state that $\vec{\nabla} f \perp \vec{t}$
- Notice that, by definition, $\vec{\nabla} g \perp \vec{t}$
- So we are simply saying that

$$\vec{\nabla} f \parallel \vec{\nabla} g \rightarrow \vec{\nabla} f = \lambda \vec{\nabla} g$$

for some scalar λ.

Aside (optional): example



Aside (optional): many variables, example of probabilities

Discrete case: Find extremum of entropy

$$S = -\sum_i p_i \ln p_i$$

with constraints

$$\sum_i p_i = 1 \quad \sum_i p_i E_i = \langle E \rangle$$

Using Lagrange multipliers:

$$\mathcal{L} = -\sum_i p_i \ln p_i - \lambda_1 \left(\sum_i p_i - 1 \right) - \lambda_2 \left(\sum_i p_i E_i - \langle E \rangle \right)$$

Setting $\frac{\partial \mathcal{L}}{\partial p_i} = 0$:

$$-\ln p_i - 1 - \lambda_1 - \lambda_2 E_i = 0 \quad \Rightarrow \quad p_i = \exp(-\lambda_2 E_i - \lambda_1 - 1)$$

This is the **Boltzmann distribution**: $p_i \propto \exp(-\beta E_i)$ with $\beta = \lambda_2$.

Aside (optional): Functional Derivatives

- Functional derivative is the extension of ordinary derivative to functionals (functions of functions)
- A general formulation (that you may have already seen) starts from an **action functional** $S[\phi]$:

$$S[\phi] = \int \mathcal{L}(\phi(x), \phi'(x), x)dx$$

$\mathcal{L}$ is the **Lagrangian function**: e.g. x is space, $\phi(x)$ a field defined over space (e.g. density)

- Perturb $S[\phi]$ by an amount ϵ with an arbitrary function $\eta(x)$ vanishing at the boundaries

$$S[\phi + \epsilon\eta] = \int \mathcal{L}(\phi + \epsilon\eta, \phi' + \epsilon\eta', x)dx$$

Expanding to first order in ϵ:

$$S[\phi + \epsilon\eta] \approx S[\phi] + \epsilon \int \left[\frac{\partial \mathcal{L}}{\partial \phi}\eta + \frac{\partial \mathcal{L}}{\partial \phi'}\eta'\right] dx$$

Aside (optional): Functional Derivatives

Integrating by parts the second term:

$$\int \frac{\partial \mathcal{L}}{\partial \phi'} \eta' dx = \left[\frac{\partial \mathcal{L}}{\partial \phi'} \eta \right]_{\text{boundary}} - \int \frac{d}{dx} \left(\frac{\partial \mathcal{L}}{\partial \phi'} \right) \eta dx$$

Since $\eta(x)$ vanishes at the boundaries, the first term drops out, giving:

$$\delta S = \int \left[\frac{\partial \mathcal{L}}{\partial \phi} - \frac{d}{dx} \left(\frac{\partial \mathcal{L}}{\partial \phi'} \right) \right] \eta(x) dx$$

If we want $S[\phi]$ to be stationary with respect to arbitrary perturbations $\eta(x)$, then $\delta S = 0$ for all $\eta(x)$.

This leads to the **Euler-Lagrange equation**:

$$\frac{\partial \mathcal{L}}{\partial \phi} - \frac{d}{dx} \left(\frac{\partial \mathcal{L}}{\partial \phi'} \right) = 0$$

which defines the functional derivative as

$$\frac{\delta S}{\delta \phi(x)} = \frac{\partial \mathcal{L}}{\partial \phi} - \frac{d}{dx} \left(\frac{\partial \mathcal{L}}{\partial \phi'} \right)$$

Aside (optional): continuous limit

Replace sums with integrals. Let $p_i \to p(x)$ where x is a continuous variable.

Constraints become:

$$\int p(x)dx = 1, \quad \int p(x)E(x)dx = \langle E \rangle$$

Entropy functional:

$$S[p] = -\int p(x) \ln p(x) dx$$

Lagrangian:

$$\mathcal{L} = -\int p(x) \ln p(x) dx - \lambda_1 \left(\int p(x)dx - 1 \right) - \lambda_2 \left(\int p(x)E(x)dx - \langle E \rangle \right)$$

Functional derivative $\frac{\delta \mathcal{L}}{\delta p(x)} = 0$:

$$-\ln p(x) - 1 - \lambda_1 - \lambda_2 E(x) = 0 \quad \Rightarrow \quad p(x) = \exp(-\beta E(x) - c)$$

Maier-Saupe Theory of Nematic Liquid Crystals

- We saw the Landau-de Gennes free energy for nematic liquid crystals.

$$f - f_0 = \frac{a}{3}\left(T - T^*\right) S^2 - \frac{2b}{27} S^3 + \frac{c}{9} S^4.$$

  - phenomenological (the coefficients are results of a perturbation)
  - based on symmetries only (rotational invariance, head-tail symmetry)
  - no molecular basis
- The **Maier-Saupe theory** is a simple alternative **microscopic** mean-field theory that gives us a more physical insight.
- The idea is to construct the free energy as we have done in various prior cases by identifying:
  - an entropic contribution
  - an energetic contribution

As for the Landau-de Gennes theory, we will focus on :

- angular deviations θ from the director and their probability distribution $p(\theta)$
- a scalar orientational order parameter $\mathcal{S}$

Maier-Saupe Theory of Nematic Liquid Crystals

Energetic Contribution

- We assume molecules interact with an angular potential that depends on their orientations via a the second Legendre polynomial $P_2(x) = \frac{1}{2}\left(3x^2 - 1\right)$:

$$V\left(\Omega_i, \Omega_j\right) = -uP_2\left(\mathbf{u}_i \cdot \mathbf{u}_j\right)$$

We can average over all possible orientations of molecule j to get the mean-field potential felt by molecule i:

$$U_i^{\mathrm{mf}}(\Omega) = \rho \int V\left(\Omega, \Omega'\right) p\left(\Omega'\right) d\Omega' = -uSP_2(\mathbf{u}_i \cdot \mathbf{n})$$

where we have defined the scalar order parameter

$$S = \int P_2\left(\mathbf{u} \cdot \mathbf{n}\right) p(\Omega) d\Omega$$

and identified the director $\mathbf{n}$ and the energetic **coupling** u.

Average over all orientations of molecule i to get the average interaction energy per molecule (and divide by 2 to avoid double counting):

$$U^{mf} = -\frac{u}{2}S^2$$

Maier-Saupe Theory of Nematic Liquid Crystals

Entropic contribution

The second contribution is **entropic** in nature.

Use orientational distribution function $p(\Omega)$ and (as in Landau-de Gennes) and assume uniaxial symmetry around the director $\mathbf{n}$, i.e. $p(\Omega) \sim p(\theta)$.

The usual Gibbs (or Shannon) entropy functional $-k_B \int p(\Omega) \ln p(\Omega) d\Omega$ contributes to the free energy as

$$f_{\text{entropy}} = k_B T \int_0^\pi p(\theta) \ln(2p(\theta)) \sin\theta d\theta$$

where $\sin\theta d\theta$ comes from the solid angle and the factor 2 comes from the normalisation $\int_0^\pi p(\theta) \sin\theta d\theta = 1$. [1]

1. Check that for the isotropic distribution $p(\theta) = 1/2$ the entropy term is zero.

Maier-Saupe Theory of Nematic Liquid Crystals

Together we, have the Maier-Saupe free energy functional per molecule

$$f = f_0 - u\frac{S^2}{2} + k_B T \int_0^\pi p(\theta) \ln(2p(\theta)) \sin\theta d\theta$$

The only constraint is normalization of $p(\theta)$:

$$\int_0^\pi p(\theta) \sin\theta d\theta = 1$$

where the $\sin\theta$ comes from the solid angle element.

Introduce the Lagrange multiplier λ to enforce the constraint and minimize the functional and set the variation to zero:

$$\delta \left[f + \lambda \left(\int_0^\pi p(\theta) \sin\theta d\theta - 1 \right) \right] = 0$$

The functional derivative yields (check!):

$$\frac{\delta f}{\delta p(\theta)} + \lambda \sin\theta = 0$$

Maier-Saupe Theory of Nematic Liquid Crystals

Remember that - by **definition** -

$$S = \langle P_2(\cos\theta) \rangle = \int_0^\pi P_2(\cos\theta) p(\theta) \sin\theta d\theta$$

So

$$\frac{\delta S^2}{\delta p} = 2 S P_2(\cos\theta) \sin\theta$$

So when calculating the partial derivative we have

$$\frac{\delta f}{\delta p} = -uSP_2(\cos\theta)\sin\theta + k_B T \left(\ln(2p(\theta)) + 1\right)\sin\theta$$

and hence, after simplifying $\sin\theta$, our Lagrange equation reads

$$-uSP_2(\cos\theta) + k_B T(\ln(2p(\theta)) + 1) + \lambda = 0$$

Its solution yields

$$p(\theta) = \frac{1}{Z} \exp\left(\frac{uS}{k_B T} P_2(\cos\theta)\right)$$

where the partition function Z ensures normalization.

Self-consistency in the Maier-Saupe Result

The solution is said to be **self-consistent** because $p(\theta)$ depends on S, which in turn depends on $p(\theta)$.

The self-consistency condition is

$$S = \int_0^\pi P_2(\cos\theta) p(\theta) \sin\theta d\theta$$

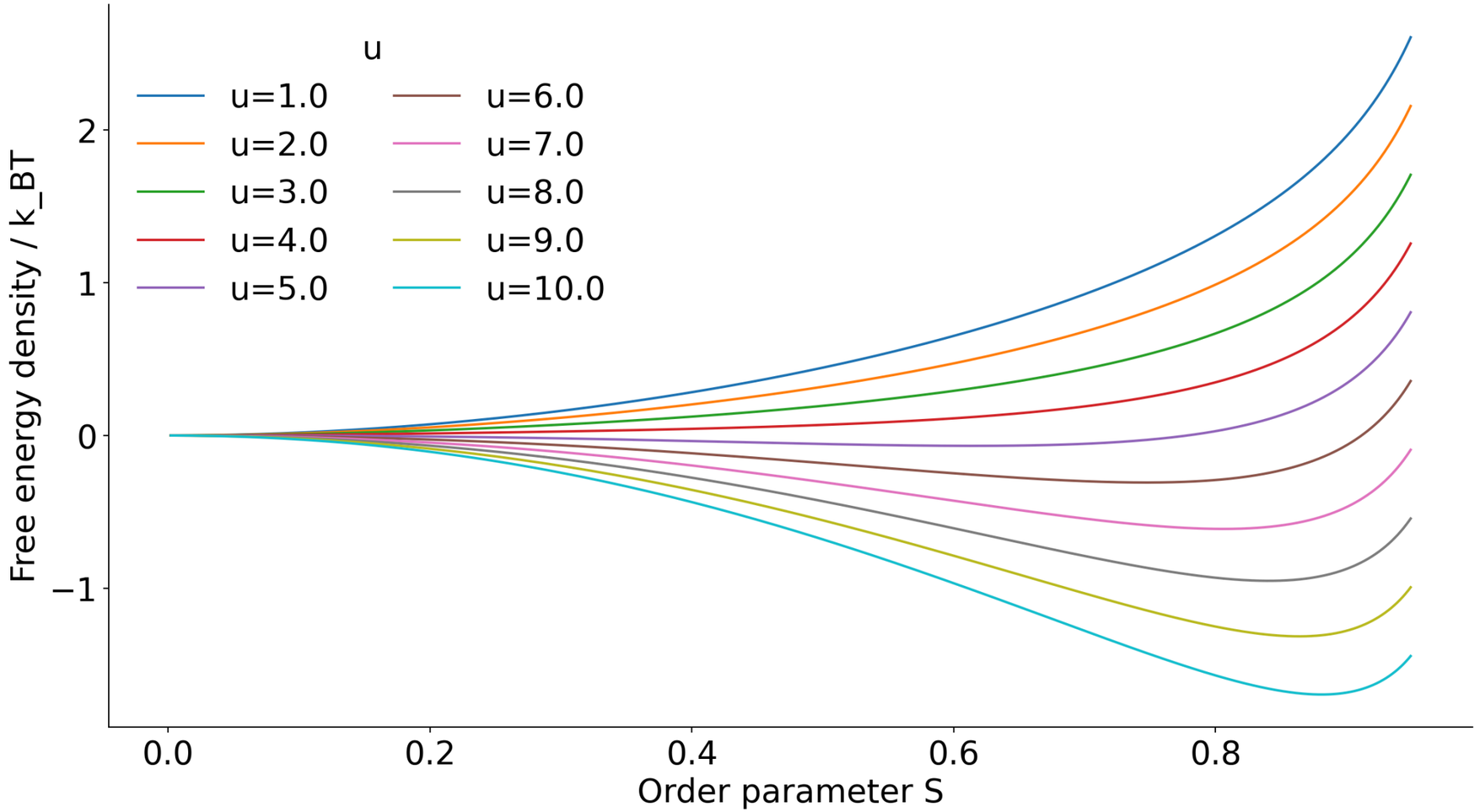
where $p(\theta)$ itself depends on S:

$$p(\theta) = \frac{1}{Z} \exp\left(\frac{uS}{k_B T} P_2(\cos\theta) \right)$$

This equation must be solved for S at each temperature T.

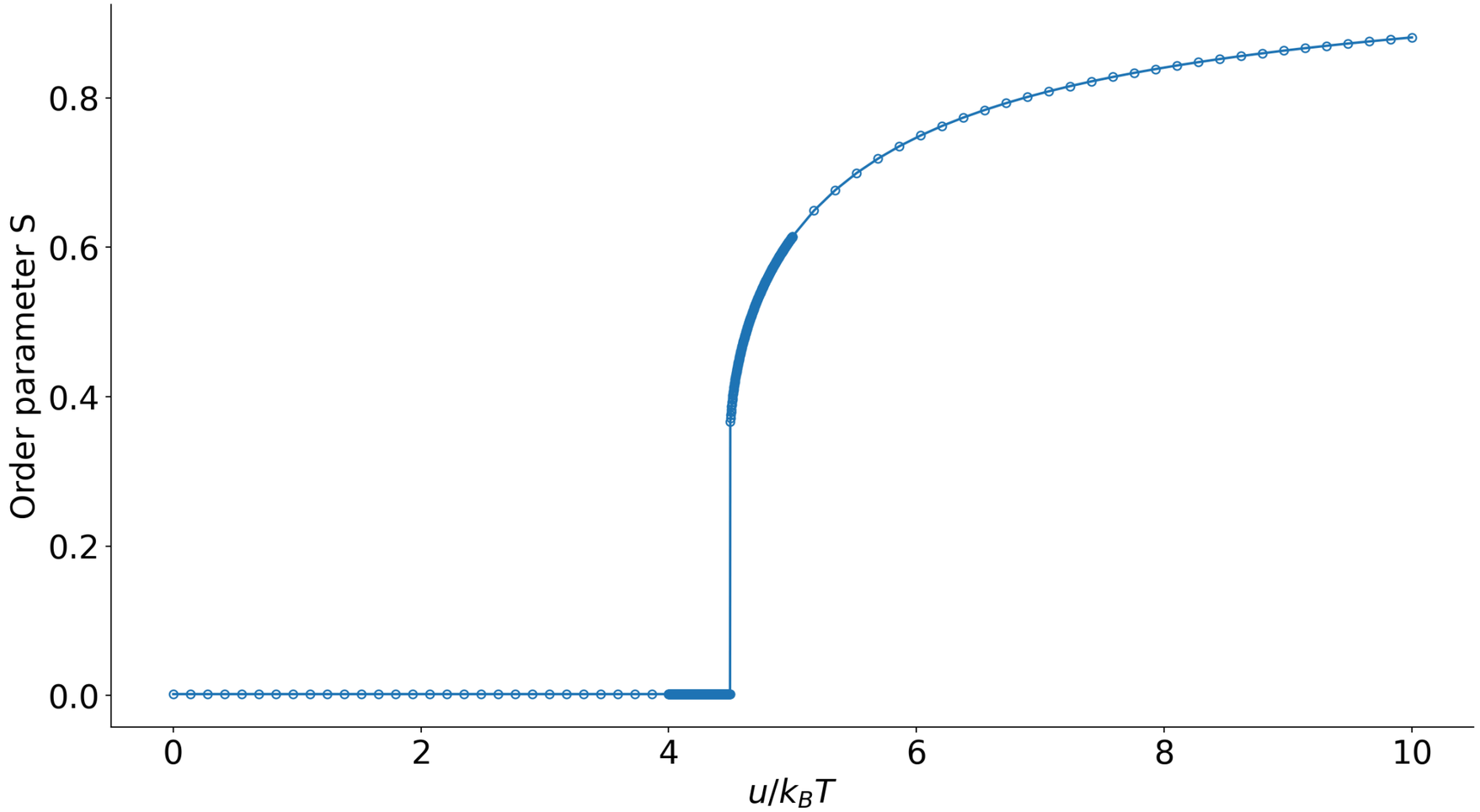
In general, this requires a **numerical** solution (see problem classes for a special case where we can make progress analytically).

Aside (optional): Maier-Saupe numerical solution



Play with the numerical solution

Aside (optional): Maier-Saupe numerical solution



The resulting phase transition is weakly first order with a discontinuous jump in S at the transition temperature. $u/k_BT \approx 4.55$.

Quick comparison Maier-Saupe vs Landau-de Gennes

- Both theories predict a first-order isotropic-nematic phase transition with a discontinuous jump in the order parameter S.
- Landau-de Gennes is phenomenological, Maier-Saupe is microscopic.
- Maier-Saupe provides a molecular basis for the coefficients in Landau-de Gennes
- Both theories can be extended to include spatial variations in the order parameter and director field, leading to descriptions of defects and elasticity in nematic liquid crystals.
- Maier-Saupe requires self-consistency requirement, while Landau-de Gennes is often simpler to analyze analytically: (slightly) harder numerics.
- Easier to cast Maier Saupe onto a lattice model (Lebwohl-Lasher).

Hard rods: ordering from entropy alone

Maier-Saupe needs $u \neq 0$. But even **hard rods**, with no attractions at all, go from **isotropic** to **nematic** at high density (Onsager, 1949): the transition is **purely entropic**.

Two entropic contributions compete:

- **orientational entropy** favours the **isotropic** phase: aligning reduces the available orientations
- **translational entropy** favours the **nematic** phase: aligned rods exclude less volume to each other, so each rod has more free volume to move in

At high density the translational gain wins: the nematic phase has the **higher** total entropy.

> **Note**
>
> Same story as the freezing of hard spheres: the spheres lose the disorder of their positions, but each gains free volume to vibrate around its lattice site.

Aside (optional): simple lattice model, Lebwohl-Lasher

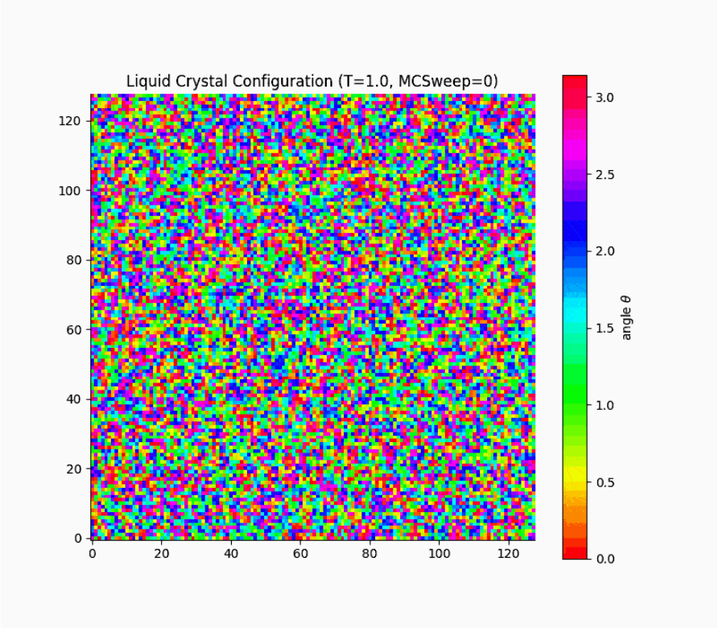
- A way to construct a Maier-Saupe-like model on lattice.
- Vectors on a lattice represent the orientations of molecules.
- Nearest-neighbor interactions favor alignment of neighboring vectors.

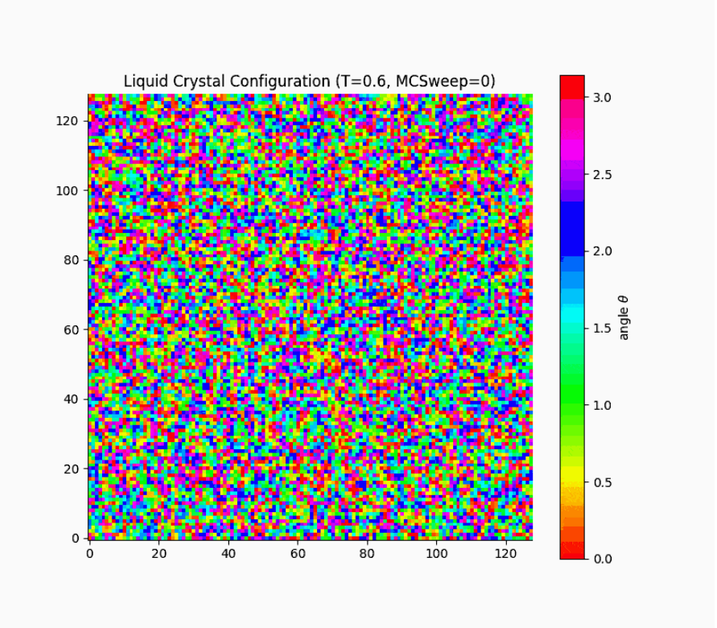
In 2D

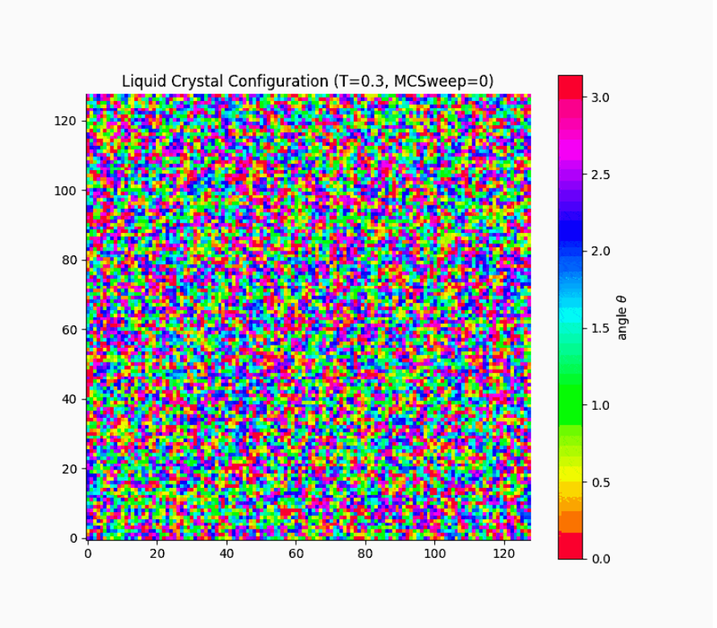
$$H = -\epsilon \sum_{\langle i,j \rangle} \left[\frac{3}{2} \cos^2 \left(\theta_i - \theta_j \right) - \frac{1}{2} \right]$$

🪁 Demonstration in class

Aside (optional): simple lattice model, Lebwohl-Lasher







in 2d the transition is **rounded**, not true first order (Kosterlitz-Thouless only quasi long range order) due to a key theorem due to **Mermin and Wagner** which states that

> **(Mermin-Wagner theorem)** Continuous symmetries **cannot be spontaneously broken** at finite temperature in systems with sufficiently short-range interactions in dimensions $d \leq 2$.

Long-range fluctuations are favoured in $d \leq 2$.

Here the continuous symmetry is the rotational invariance of the director field in the plane.

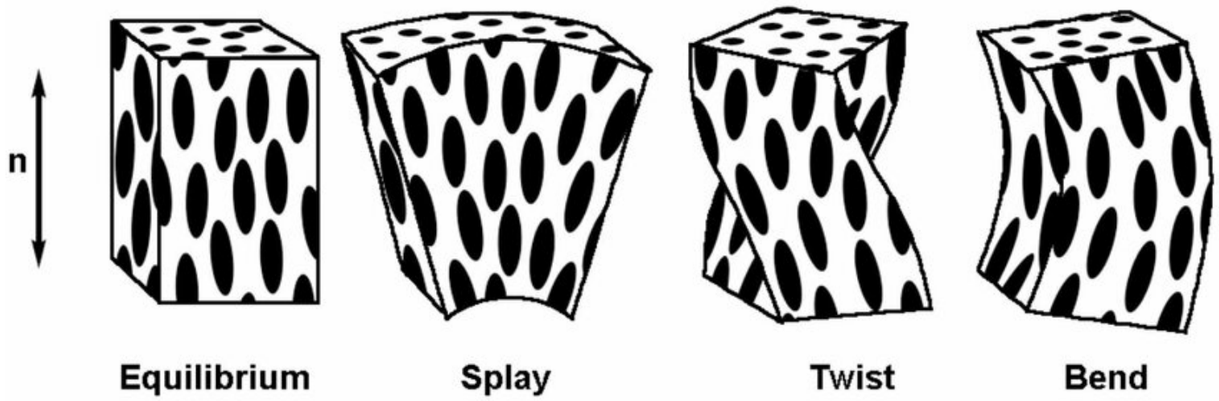
Continuum limit and Frank elasticity

Charles Frank (1958, of our Frank lecture theatre) developed a continuum theory to describe the elastic properties of nematic liquid crystals.

The Frank free energy density describes the cost of spatial variations in the director field $\mathbf{n}(\mathbf{r})$:

$$f_{\text{Frank}} = \frac{1}{2}K_1(\nabla \cdot \mathbf{n})^2 + \frac{1}{2}K_2(\mathbf{n} \cdot \nabla \times \mathbf{n})^2 + \frac{1}{2}K_3(\mathbf{n} \times \nabla \times \mathbf{n})^2$$

- K_1: **splay** elastic constant
- K_2: **twist** elastic constant
- K_3: **bend** elastic constant

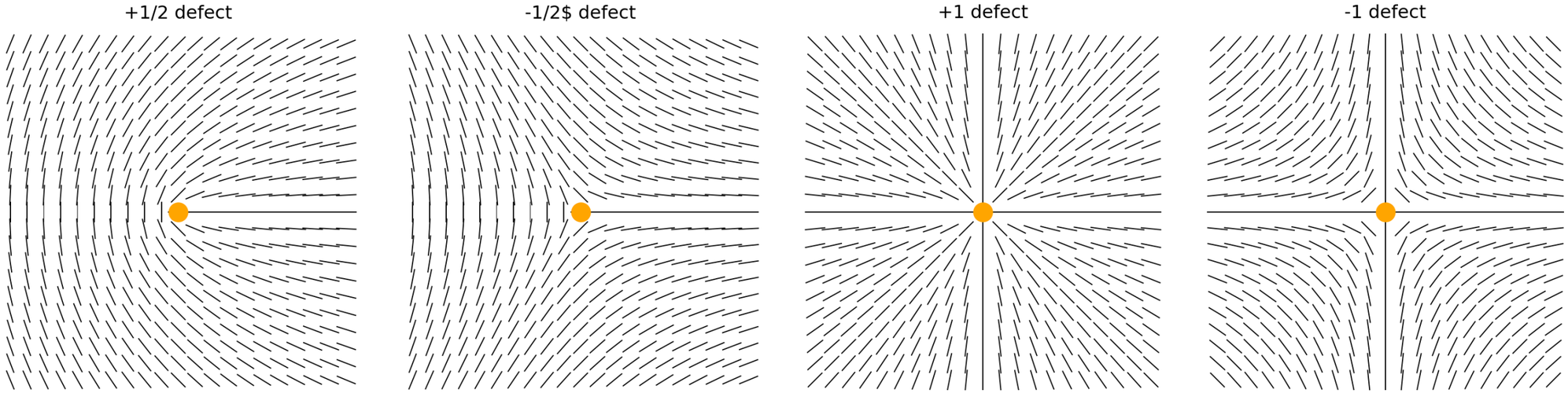


Pure splay, twist and bend compared to equilibrium from Binder et al. J. Phys. Mater. (2020)

With this free energy, one can model large-scale deformations and **defects** in nematic liquid crystals.

Topological Defects in Nematic Liquid Crystals

- We are working in the continuum with a director field $\mathbf{n}(\mathbf{r})$ that describes the local average orientation of molecules.
- In this framework, defects are **singularities** of the field, called **disclinations**
- Examples in 2D:



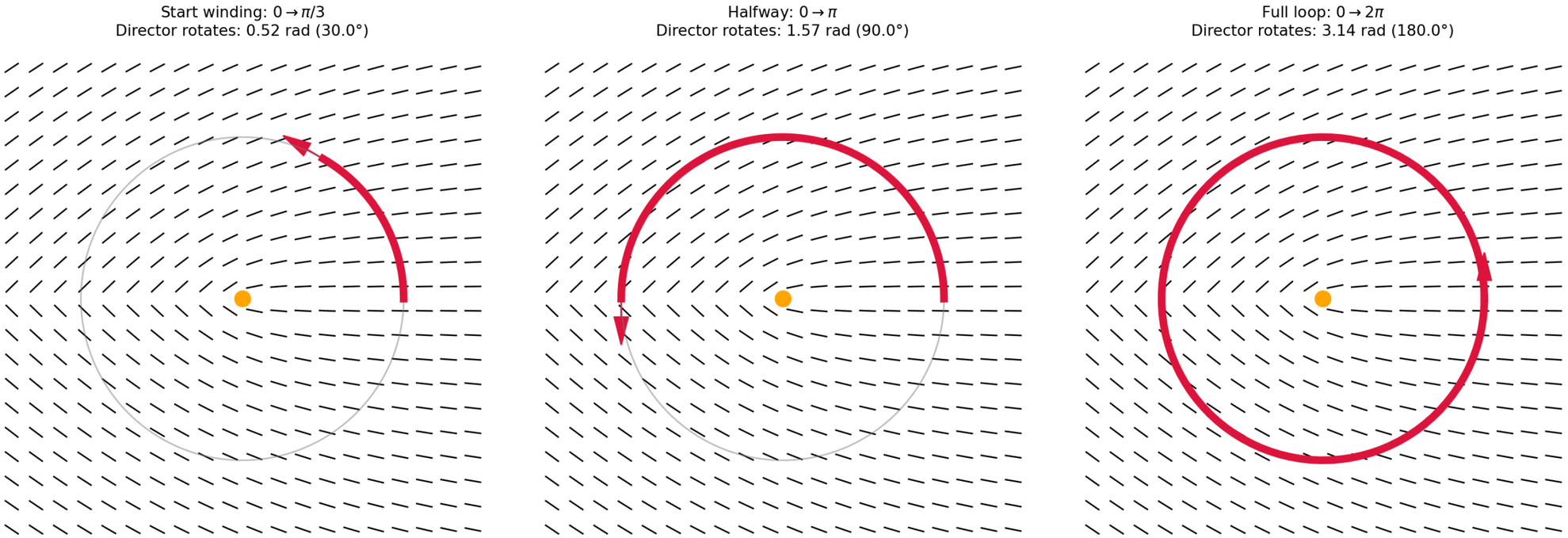
- These singularities cannot be removed by any smooth deformation of the field: they are **topologically protected**.
- To make them vanish, one needs oppositely signed defects to bring together so that they **annihilate** each other.
- At finite temperature, defects are spontaneously created/annihilated.

Topological Charge of a Defect

- The topological charge (or winding number) s of a defect quantifies how many times the director field rotates by 2π when encircling the defect once.

$$\Delta\theta = 2\pi s$$

For example, a charge of $s = 1/2$ is a rotation by π is performed when going around the defect once.



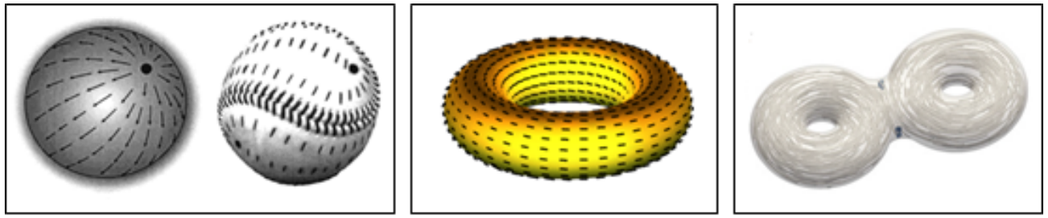
Aside (optional): conservation of topological charge

- Topological charge is **conserved**: the total charge in a closed system remains constant over time.
- Defects can be created or annihilated only **in pairs** of opposite charge (e.g., +1/2 and -1/2).
- The **topology** of the space imposes such constraints of conservation via the *Poincaré-Hopf theorem*, which states

> The total topological charge of the defects in a field confined to a compact space is equal to the Euler characteristic of that space

In practice:

- On a plane (Euler characteristic = 1), the total topological charge must sum to 1.
- On a sphere (Euler characteristic = 2), the total topological charge must sum to 2.
- On a torus (Euler characteristic = 0), the total topological charge must sum to 0. $\rightarrow$ in periodic systems the total charge must be zero.
- This imposes constraints on the dynamics and interactions of defects.

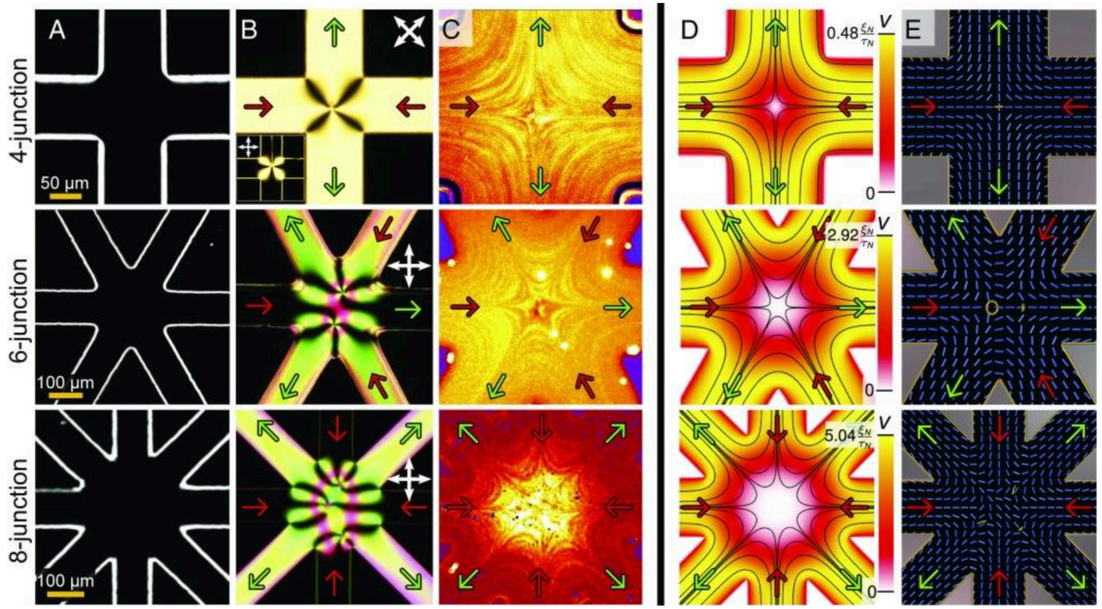


Director fields on a sphere, on a torus and a double torus. The number of handles (the genus g) determines the Euler characteristic $\chi = 2 - 2g$. From Nelson (2002)

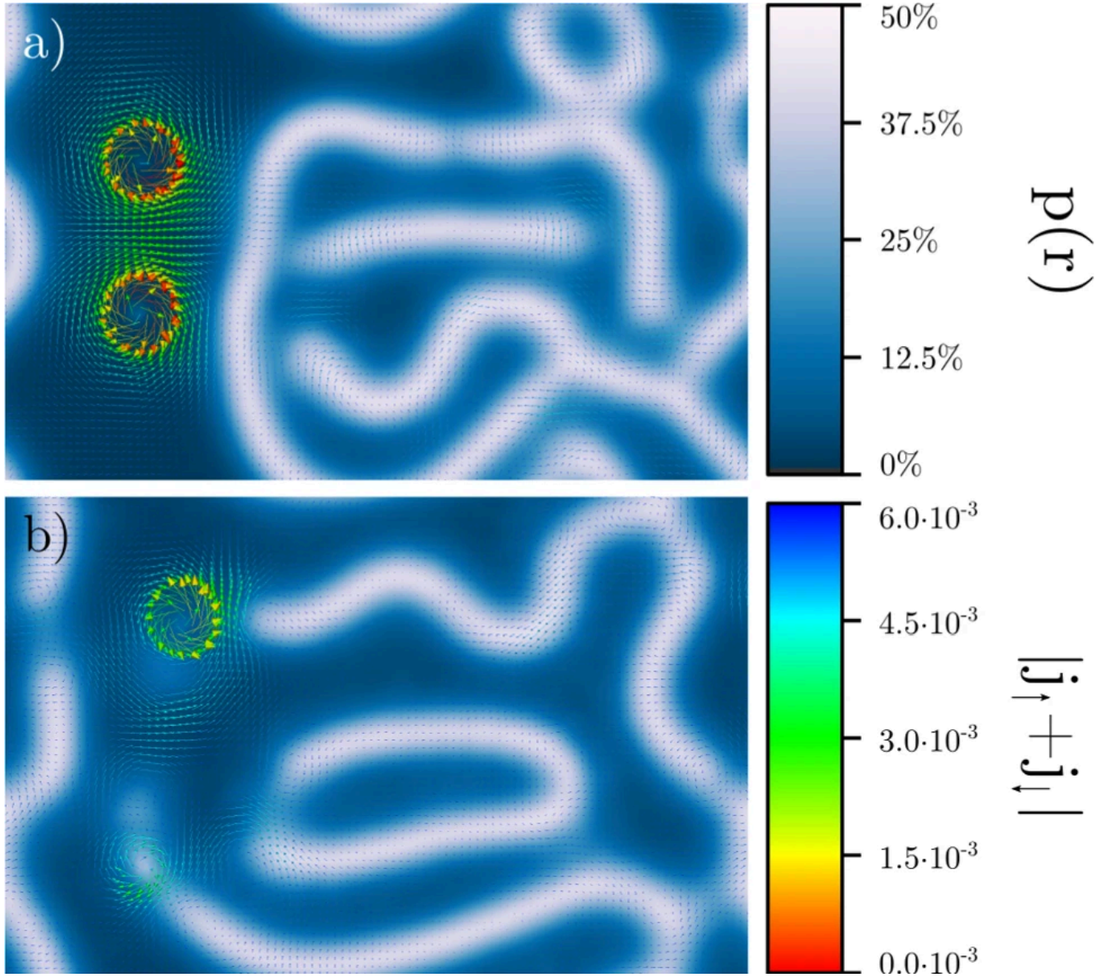
Aside (optional): topological defects

Analogies well-beyond liquid crystals:

- vortices in superfluids and superconductors
- dislocations in crystals
- electromagnetic field configurations
- flow fields microfluidic junctions



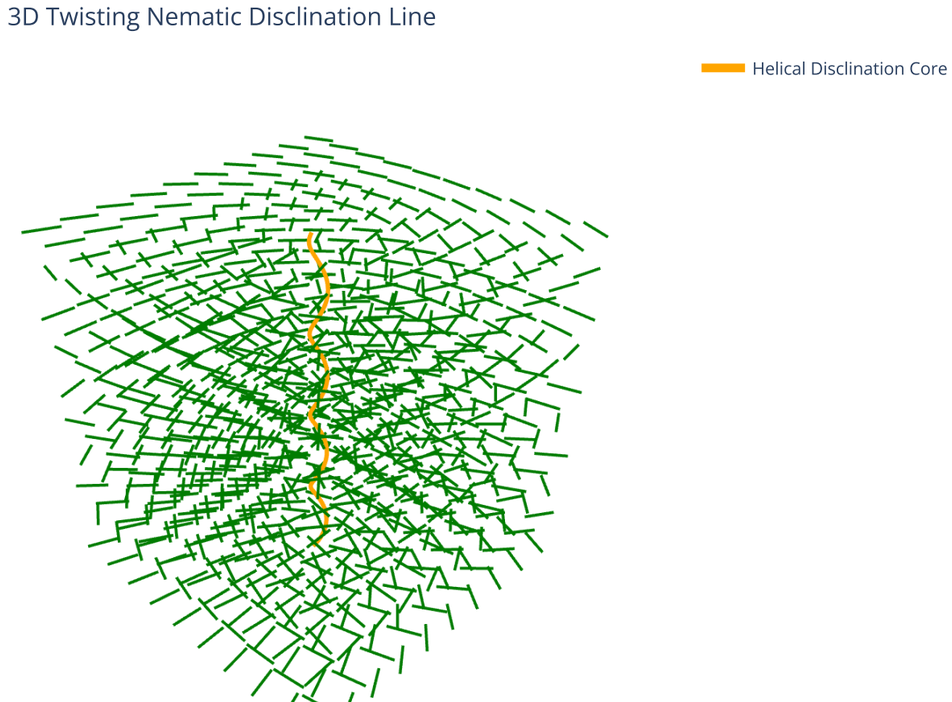
Topological defects in microfluidic devices



Vortex anti-vortex pair in a superfluid (a) initial configuration and (b) after relaxation. From Magierski, *Eur Phys J* (2024)

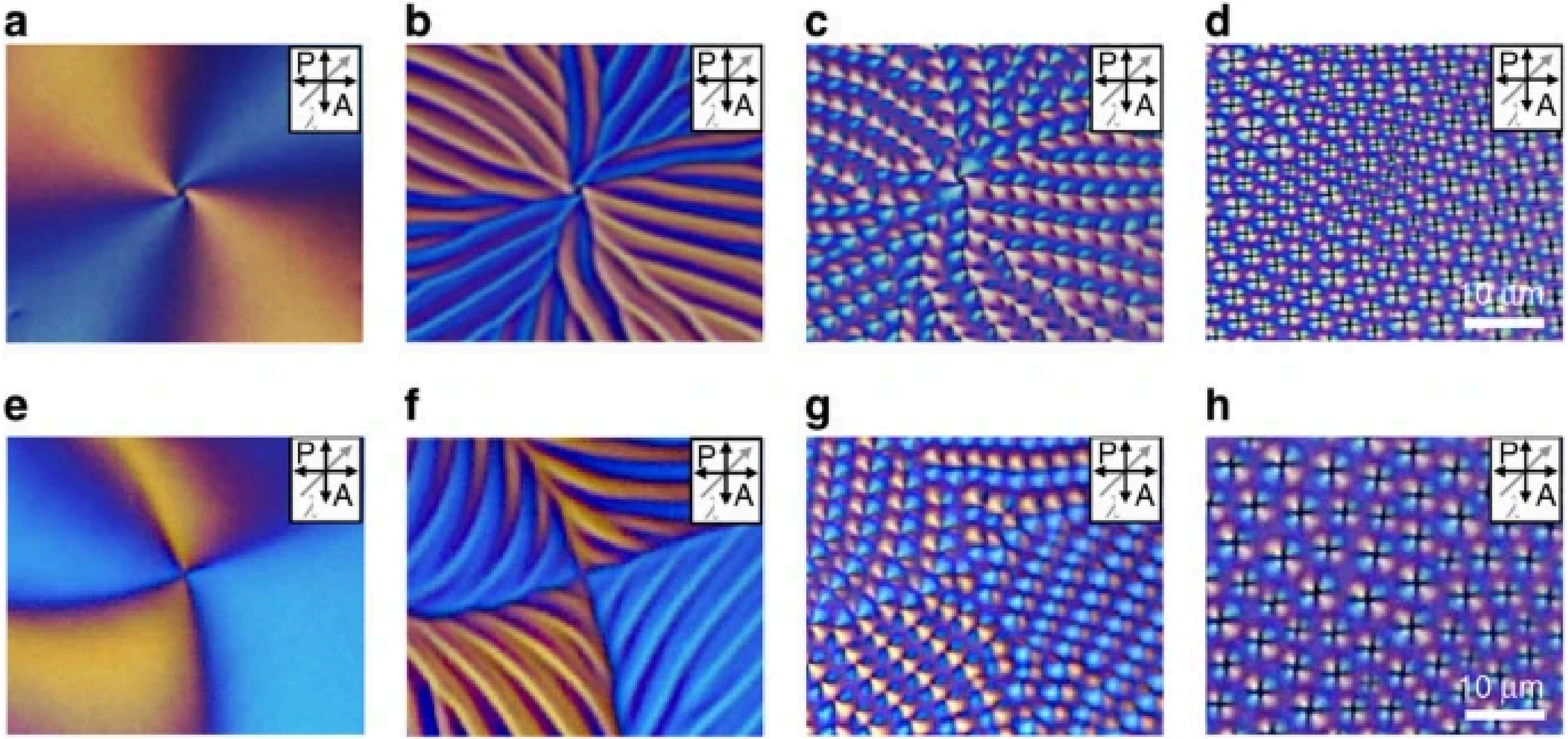
Aside (optional): defects in 3 dimensions

- In 3D the disclinations become **lines**
- The director field wraps around the line defect in a similar way as in 2D.
- The classification of disclination in 3D involves the rotation group SO(3) (that generalises the construction we have seen in two dimensions) and are more complex in general. They include
  - lines
  - closed loops
  - point-like features (hedgehogs and monopoles)
  - complex textures.



Aside (optional): defects and phase transitions

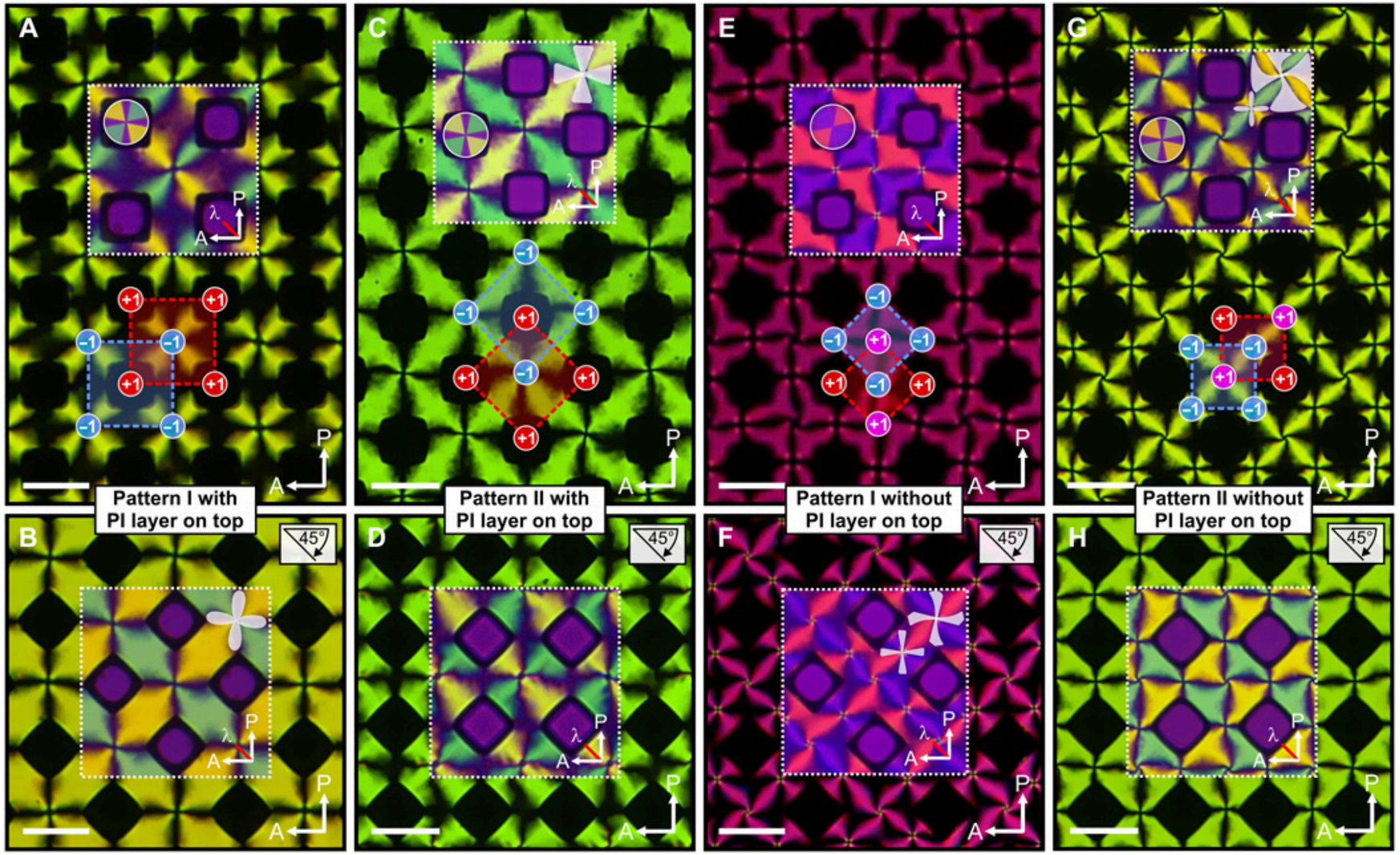
Defects control the stability and dynamics of nematic liquid crystals and drive the phase transitions between phases.



Defect formation and nematic to smectic phase transition via defect branching, Gim et al Nature Communications (2017)

Aside (optional): complex disclination patterns

Example from Kim et al Science Advances (2018)



Liquid-crystals defects created and controlled by external forces (air pillars through a substrate).

Summary: liquid crystals

Topic	Key result
Alignment tensor	$\mathbf{Q} = \frac{d}{d-1}\left\langle \mathbf{u}_i \otimes \mathbf{u}_i - \frac{1}{d}\mathbf{I} \right\rangle$: head-tail symmetry needs a tensor
Order parameter, director	$\mathcal{S}$ = largest eigenvalue of $\mathbf{Q}$, $\mathbf{n}$ = its eigenvector; 3D: $\mathcal{S} = \langle P_2(\cos\theta) \rangle$
Landau-de Gennes	$f - f_0 = \frac{a}{3}(T - T^*)\mathcal{S}^2 - \frac{2b}{27}\mathcal{S}^3 + \frac{c}{9}\mathcal{S}^4$: cubic term, first order
Maier-Saupe	$p(\theta) \propto \exp\left(\frac{u\mathcal{S}}{k_B T} P_2(\cos\theta)\right)$, self-consistent, weakly first order
Hard rods	purely entropic: orientational entropy lost, translational entropy gained
Lebwohl-Lasher	lattice Maier-Saupe; in 2D only quasi-long-range order
Frank elasticity	splay K_1, twist K_2, bend K_3
Topological defects	$\Delta\theta = 2\pi s$, $s = \pm\frac{1}{2}, \pm 1$; annihilate in $\pm$ pairs

Full summary with links in the notes